



Investigation of the influence of solid lubricants on the tribological properties of polyamide 6 nanocomposite

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ABSTRACT

The tribological properties of nano titanium dioxide filled polyamide 6 (TiO₂/PA 6 nanocomposite, 5/95 by weight) and its composites filled with single and combined solid lubricants were systematically investigated. It was found that all the solid lubricants except molybdenum disulfide (MoS₂) could significantly enhance the tribological performance of TiO₂/PA 6 nanocomposite; and the nanocomposite filled with polytetrafluoroethylene (PTFE) exhibited lower friction coefficient and wear rate than that filled with ultra-high molecular weight polyethylene (UHMWPE); but it was interesting that the nanocomposite filled with MoS₂ combined with UHMWPE had better tribological performance than that filled with MoS₂ combined with PTFE, and the nanocomposite filled with MoS₂ together with both UHMWPE and PTFE performed the best among the combination solid lubricants. XPS and SEM results showed that the synergism of fillers in helping the formation of thin, uniform and continuous transfer film was responsible for the enhancement in tribological properties.

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1. Introduction

PA 6 is a kind of important engineering plastic which is widely used in friction material field, and recently its nanocomposites have received much attention since the significant improvements in tribological properties that led by nanoparticles [1–3]. However the high surface energy of nanoparticles will usually result in irreversible aggregation, which may plow on the counterpart steel ring and lead to high friction coefficient [4].

It is well known that solid lubricants exhibit self-lubricating behavior and their applications in sliding prevent stick-slip motion instabilities. The solid lubricants such as molybdenum disulfide (MoS₂) [5–7], ultra-high molecular weight polyethylene (UHMWPE) [8], polytetrafluoroethylene (PTFE) [9,10], graphite [11] and graphene [12] have been thus widely used to enhance the friction and wear properties of polymers. Synergistic effects are usually obtained between solid lubricants which are useful for improving the tribological properties of polymer composites [13,14]. But not all cases can solid lubricants improve the tribological properties of polymers. Since solid lubricants sometimes reduce the mechanical properties of the composites, the decrease in mechanical properties will adversely affect the wear resistance [15,16]. Delightfully, this problem can be resolved by incorporating reinforcing agents such as fibers [17] and nanoparticles [18]. In this case, solid lubricants may facilitate the formation of the

transfer films while the reinforcing agents enhance the deformation resistance and plow resistance of the polymer composites, consequently the synergistic effects between different fillers are obtained which are useful for improving the tribological properties of polymer composites.

Though the reinforcing agents and the solid lubricants usually showed a positive hybrid effect [19–24] on enhancing the performance of polymers, it was not necessarily true. The tribological properties closely depended on the compositions and the tested conditions [25–27]. Whilst the synergistic effects between combined solid lubricants and TiO₂ have not systematically studied.

With those perspectives in mind, a series of PA 6 nanocomposites which used nano-TiO₂/PA 6 (5/95 by weight) as polymer matrix, and different combinations of MoS₂, UHMWPE and PTFE as fillers were prepared and their tribological properties were systematically investigated. The objective of this work is to discuss the synergism of the multiple fillers in the composite on the improvement of tribological performance.

2. Experimental

2.1. Materials

TiO₂/PA 6 nanocomposite (5/95 by weight) was used as matrix, the model of PA 6 in the composite were PB1006 LM BN 70745; the nano-TiO₂ (RCL-69) particles in the composite had an average diameter of 1–37 nm. The average size of Molybdenum disulfide

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(MoS₂) was 10 μm and PTFE was less than 1 μm. The average molecular weight of UHMWPE was 3–6 million.

2.2. Specimens preparation

The blend of TiO₂/PA 6 and its composites with solid lubricants were achieved by using twin-screw extruder ($\phi=35.5$ mm and $L/D=41$) with a screw frequency of 360 rpm and a feeding frequency of 20 Hz under processing temperatures of 210, 220, 230, 240, 240, 240, 240 and 240 °C in nine zones of the extruder barrel. The extrudate was continuously cooled by water and pelletized. The granule was dried in a vacuum oven at 60 °C for 12 h again and subsequently injection molded (on a HTF160J/TJ injection machine) to tribological test samples with a dimension of $25 \times 25 \times 15$ mm³ according to the size of specimen clamp on the tribometer. The injection pressure was 85 MPa, the dwell pressure

20 MPa and the dwell time 4 s. The temperatures in four zones were 250, 250, 250, and 250 °C and in the nozzle 230 °C.

2.3. Wear test

Friction and wear test were carried out on an UMT-3 (Universal Macro Materials Tester, Fig. 1) ball-on-flat tribometer. An assembly diagram of the friction pairs was shown in Fig. 2, the ball up specimen (HRC 62) with a diameter of $d=9.5$ mm used as counterface, consisted of 440-C stainless which contains about 17.5% chromium. The load was applied downward through the ball counterface against the flat tested specimen mounted on a reciprocating drive. The surface were cleaned ultrasonically with acetone and thoroughly dried before testing. All the test were performed in normal laboratory environment (temperature: 20 ± 5 °C, humidity: $50 \pm 10\%$) with four loads: 40 N, 60 N, 80 N and 100 N and at four velocities 200 rpm, 500 rpm, 1000 rpm and 1500 rpm (one motion period of forward and backward is defined as one round, and the stroke is 10 mm) which were based on the use condition (low load and low velocity) of the products. The test duration ranged from 0 min to 120 min. The friction coefficient was recorded and calculated by the ratio between the tangential force (F_x) and normal load (F_z), which obtained directly from the equipment. The average values of friction coefficient in the test range were used as the friction coefficient of samples. The width of the wear track b (see Fig. 3) was measured with a KH-7700 digital microscope to an accuracy of 0.001 μm. Then the wear rate was calculated using the following equation:

$$\text{Wear rate} = \frac{\Delta v}{Ld} = \frac{B}{Ld} \left[\frac{\pi r^2}{180} \sin^{-1} \left(\frac{b}{2r} \right) - \frac{b}{2} \sqrt{r^2 - \frac{b^2}{4}} \right] [\text{mm}^3/\text{Nm}]$$

where B is the trace of friction (10 mm), r is the semi diameter of the chromium steel ball (mm), b the width of the wear trace (mm), L the load (N) and d the sliding distance (m).

Each test was repeated three times, the maximum variation between these experimental values was controlled within $\pm 15\%$. The data represented in this paper were the arithmetic mean values of the tests.

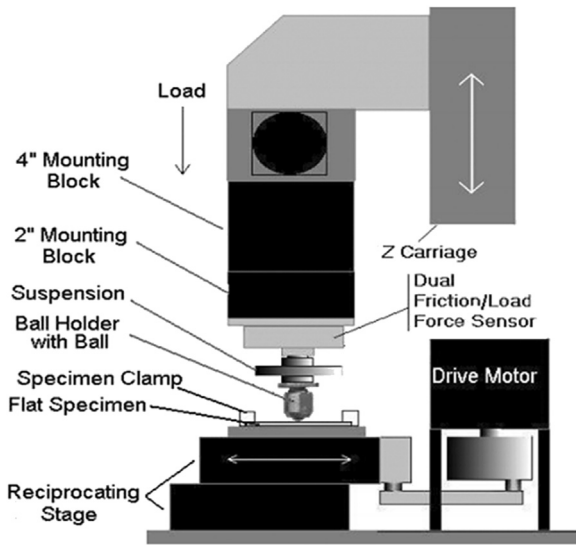


Fig. 1. Schematic diagram of ball-on-flat tribometer.

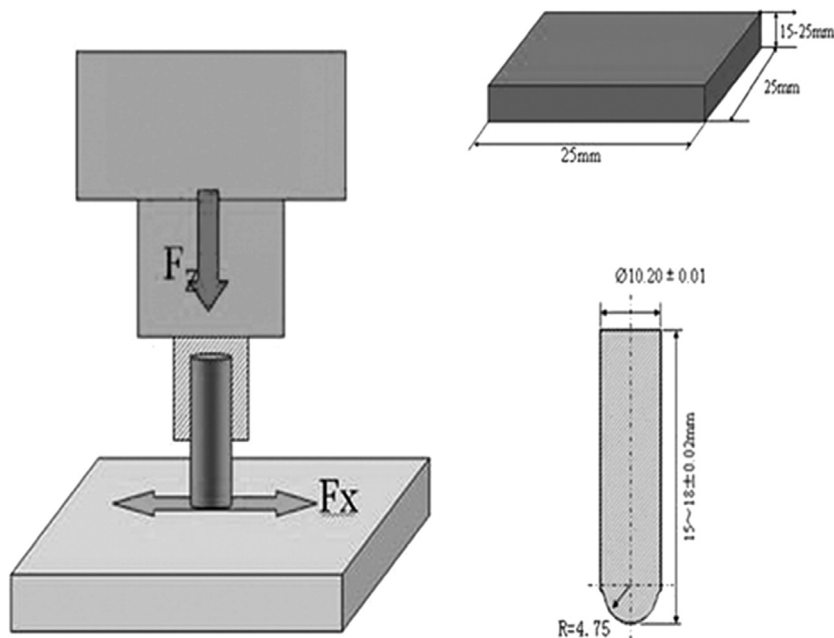


Fig. 2. Schematic diagram and sample size for the frictional couple F_z -load direction; F_x -friction direction.

The morphologies and element chemical states of the worn surface and transfer film on the counterface were examined with scanning electron microscope (Nova NanoSEM 230) and X-ray photoelectron spectroscopy (XPS), respectively.

3. Results and discussion

3.1. Friction and wear properties

The friction coefficient and the wear rate of nano titanium dioxide filled polyamide 6 (TiO_2/PA 6 nanocomposite, 5/95 by weight) and its composites filled with different solid lubricants at 40 N and 200 rpm are shown in Fig. 4. It is clearly seen that UHMWPE or PTFE as fillers can significantly reduce the friction coefficient and wear rate, but MoS_2 as filler is harmful to the improvement of the tribological properties of TiO_2/PA 6 nanocomposite. The best result is achieved for 5 wt% PTFE filled composite, the friction coefficient of which is 0.086 and the wear rate is $9.8 \times 10^{-6} \text{ mm}^3/\text{Nm}$ while that of the matrix is 0.544 and $4.35 \times 10^{-4} \text{ mm}^3/\text{Nm}$, respectively. Comparing the independent effect on the friction and wear of these three fillers, PTFE is the most effective one; the next is UHMWPE; while MoS_2 exhibits the

negative effect; it is also clear that the effect of the kind of solid lubricants is more obvious than the content. The improvements can attribute to the lubricating capability of the solid lubricants; UHMWPE or PTFE can embody the agglomerated nanoparticulates and reduce the abrasive wear. Also the nanoparticles embed into the friction surface make the friction surface strengthened and recoverable. Nanoparticles are also known to facilitate the formation of transfer film and strengthen the interaction between transfer film and the counterpart [18]. The synergistic effects of nanoparticles and solid lubricants are responsible for the enhancements in tribological properties.

Based on the above results, it is decided to combine MoS_2 with UHMWPE and/or PTFE, and then use the combinations as respective fillers in the hope of synergism between the solid lubricants. The comparisons of the friction coefficient and wear rate between TiO_2/PA 6 nanocomposite and its composites that filled with a single phase of solid lubricants and with multiple phases of solid lubricants in a constant content of 15 wt% are given in Fig. 5. The results clearly show that the composite filled only with 15 wt% PTFE shows the best tribological properties. In addition to the lubrication effects of solid lubricants, maybe the unique fibrillation response to stress concentration at a sharp crack of PTFE (in the presence of nanoparticles) is the other important reason of the lowest friction coefficient and wear rate [28]. Easy fibrillation at surface crack tips may interrupt crack propagation, substantially improve the mechanical properties of the near surface polymer and enhance the ability to resist deformation, subsequently the tribological properties were improved. From Fig. 5 it can be also seen that the combined solid lubricants decrease the friction coefficient and wear rate of the composite filled with MoS_2 alone remarkably, and meanwhile the friction coefficient and wear rate of the composites with combined solid lubricants even much lower than that of TiO_2/PA 6 nanocomposite matrix; The results are interesting, since the composition with both MoS_2 and UHMWPE exhibits better tribological properties than the one with both MoS_2 and PTFE, although PTFE shows better results than UHMWPE as a single phase of filler. Further, the combination of UHMWPE and MoS_2 shows synergistic effect; the friction coefficient and wear rate of the composite filled with both UHMWPE and MoS_2 are lower than that of the composites containing 15 wt% UHMWPE or MoS_2 alone. A reasonable explanation can be analyzed as follows: the melting point of UHMWPE is lower than PTFE, as the friction proceeding, the UHMWPE melts due to the friction induced heat effects, then plays a role of adhesion agent. So other fillers are not easily chipped off from the friction surface

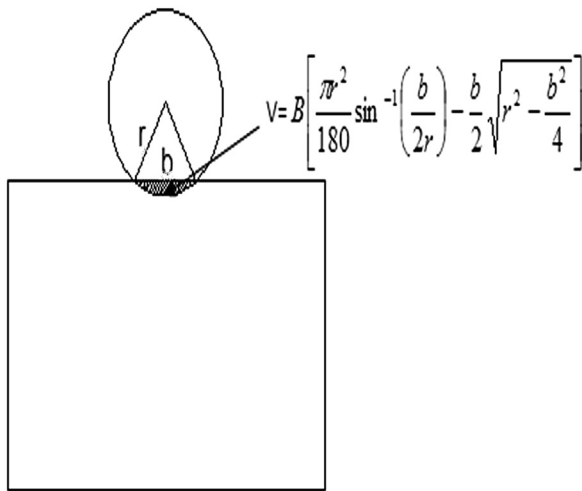


Fig. 3. Calculation of wear volume (V).

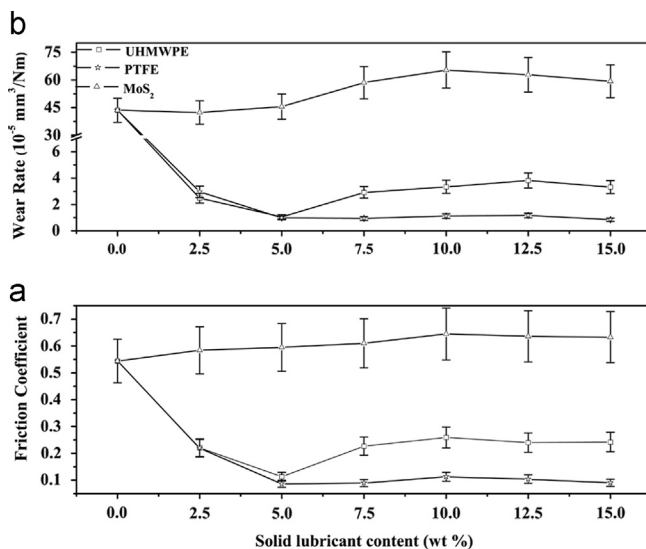


Fig. 4. Effect of solid lubricant contents on (a) the friction coefficient and (b) wear of TiO_2/PA 6 nanocomposite and its composites at 40 N and 200 rpm.

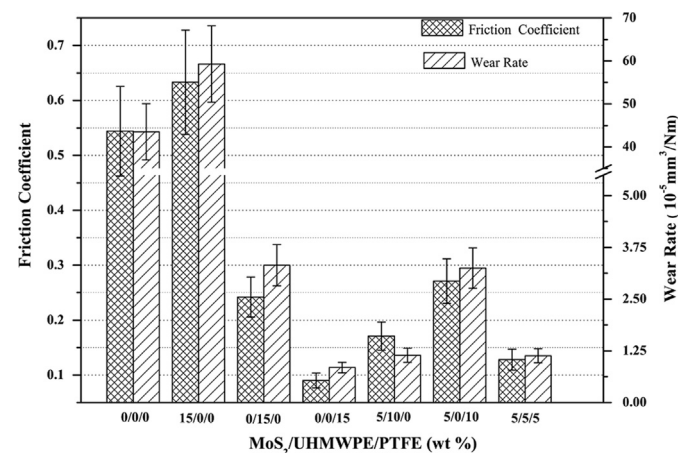


Fig. 5. Comparisons of the friction coefficient and wear rate of TiO_2/PA 6 nanocomposite and its composites filled with various solid lubricants at 40 N and 200 rpm. The total solid lubricant content is constant of 15 wt%.

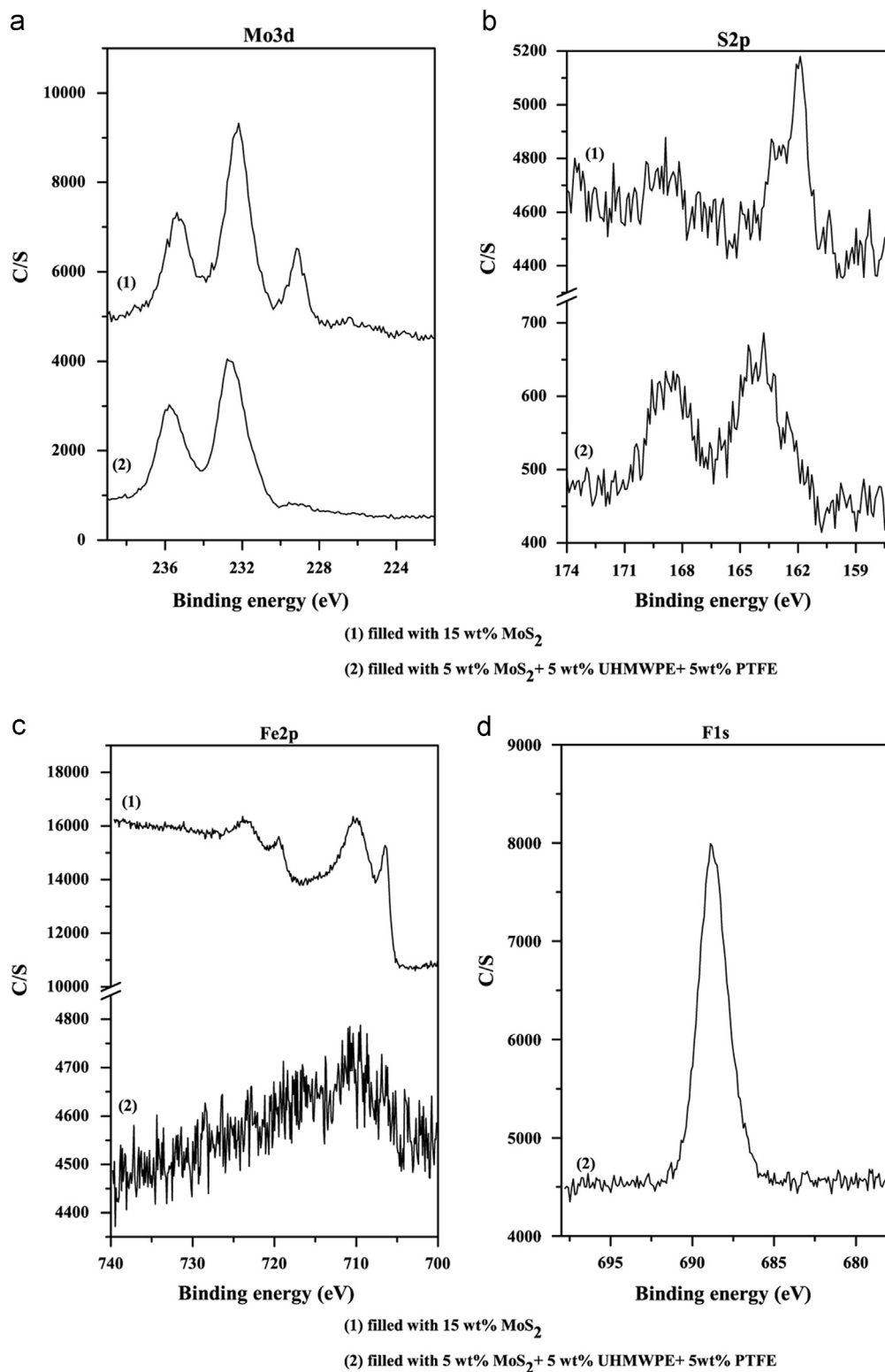


Fig. 6. XPS spectra of the transfer film formed on the counterface by TiO₂/PA 6 nanocomposite filled with (1) 15 wt% MoS₂ and (2) 5 wt% MoS₂+ 5 wt% UHMWPE+ 5 wt% PTFE.

and take a full play in enhancing the tribological properties. Subsequently, the friction and wear could be reduced.

Fig. 5 also shows the results of the compositions with a constant amount of 5 wt% MoS₂ and various amounts of UHMWPE and PTFE, among these combined solid lubricants filled composites the best result is demonstrated by the composition with 5 wt% MoS₂+ 5 wt% UHMWPE+ 5 wt% PTFE, possessing a friction

coefficient of 0.128 and a wear rate of $1.13 \times 10^{-5} \text{ mm}^3/\text{Nm}$, which is about 40 times lower than the value measured for the TiO₂/PA 6 nanocomposite matrix. The tribological properties could be improved when fillers or matrix decomposed and then generated reaction products which enhanced the bonding between the transfer film and the counterface [16]. The XPS results of the transfer films, as shown in Fig. 6, formed by the TiO₂/PA 6 nanocomposites filled with

15 wt% MoS₂ and with 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE rubbing against a steel ball have proved this point. For MoS₂ single phase filled TiO₂/PA 6 nanocomposite, the S2p peak at 162.5 eV corresponds to S in MoS₂, the Mo3d peak at 229.5 eV confirms this point. The Mo3d peak at 232.7 is attributed to Mo in MoO₃. Fe in the steel ball is indicated by the peak at 706.9 eV. The peaks at 709.7–710.6 eV indicate the presence of FeO and Fe₂O₃. In summary, in the case of MoS₂ as filler, a large proportion of MoS₂ had not been involved in chemical reactions; partial proportion of MoS₂ and substrate iron had been oxidized into MoO₃ and Fe₂O₃. It has been reported elsewhere [17] that MoO₃-nylon was easily transferred to the counterface, but the adhesive strength of the transfer film was too weak to resist the rubbing. This may correspond to the high friction coefficient and wear rate of MoS₂ filled TiO₂/PA 6 nanocomposite. In the case of multiple phases of fillers (5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE) filled TiO₂/PA 6 nanocomposite, MoS₂ had not been found in the transfer film, because all of the MoS₂ had already been oxidized into MoO₃. And the active S reacted with Fe in the counterface so that FeS, FeSO₄, and Fe₂(SO₄)₃ would be formed. The peaks at 161.6, 168.9 and 169.2 eV in the S2p spectrum correspond to S in FeS, FeSO₄, and Fe₂(SO₄)₃, respectively. The broad peak of Fe2p from 710.6–712.4 eV also demonstrates the presence of iron sulfide. The F1s spectrum shows only one peak at 689.1 eV that corresponds to F in PTFE, indicating that PTFE did not undergo any tribological reaction during sliding. In short, the presence of UHMWPE and PTFE helps in some ways in promoting the formation of FeS, FeSO₄, and Fe₂(SO₄)₃, which increase the adhesion between the transfer film and the counterface, therefore the friction coefficient and wear rate reduced.

The effect of mass ratio of MoS₂/UHMWPE on the friction coefficient and wear rate of TiO₂/PA 6 nanocomposite at 40 N and 200 rpm is shown in Fig. 7, the total weight content of solid lubricants was fixed at 15 wt%. It is observed that the combined solid lubricants decrease the friction coefficient and wear rate of TiO₂/PA 6 nanocomposite remarkably; with the decrease of MoS₂ and the increase of UHMWPE in the lubrication system, the friction coefficient and wear rate decrease firstly then increase when met an optimum ratio, the lowest friction coefficient (0.17) and wear rate ($1.14 \times 10^{-5} \text{ mm}^3/\text{Nm}$) can be achieved as the ratio of MoS₂/UHMWPE is 5/10, and they are approximately reduced by a factor of 3 and 40 than that of TiO₂/PA 6 nanocomposite. The results reveal that both MoS₂ and UHMWPE play an important role in the lubrication system. As mentioned before, if the MoS₂ content in the solid lubrication system is too high, MoS₂ will be oxidized and generate the MoO₃ but few iron sulfides which will affect the lubrication property and the adhesion

strength to the metal surface of MoS₂. But if the UHMWPE content in the solid lubrication system is too high (MoS₂ too little), in spite of the UHMWPE can fully contact with the counterface, it will not produce obvious chemical bond such as MoS₂ into iron sulfides as the previous section illustrated, so the friction and wear rate will be still high. At an optimum proportion of MoS₂/UHMWPE and with the repetitive sliding, MoS₂ and nano-TiO₂ may be stuck by the melted UHMWPE, and not easily to be worn, as a result the nano-TiO₂ can support the bulk of the normal load. So high flash temperature will be generated in the contacting filed [29], this may be beneficial to the reaction between MoS₂ and the substrate iron to produce iron sulfides which are responsible for the good tribological performances. So it can be concluded that the synergy of solid lubricants has been given full play at a proper proportion of MoS₂/UHMWPE that is 5/10.

Fig. 8 shows the friction and the wear behavior of TiO₂/PA 6 nanocomposite filled with 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE as a function of load and velocity. When the load varied from 40 N to 100 N, the velocity was fixed at 200 rpm; and when the velocity varied from 200 rpm to 1500 rpm, the load was fixed at 40 N. It is clearly that the effect of load and velocity on the friction coefficient is not obvious according to the scope of experimental error. It may be due to the frictional heat which will raise the temperature of the friction surfaces, therefore decrease the shear strength of the material, as well as increase the real contact area. These two aspects will affect the friction coefficient competitively; the final friction coefficient of the polymer will be determined by the dominant aspect. In this case, the effects of these two competitive aspects are seemed to be equal, so the friction coefficient is virtually not affected by load and velocity. It can also be seen that the wear rate increases with load which is similar to the results reported elsewhere [4,27], but decreases with the increase of velocity, which can be attributed to the friction-induced heat effects. At higher load and velocity, the temperature of the worn surface rose, the sample surface was intenerated and viscous flow occurred. Subsequently, the tendency of conglutination and the transfer of matrix increased while the adhesive power decreased. Consequently, the formation of transfer film seemed to be more difficult, and the materials were rubbing off easily, resulting in higher wear loss. As shown in Fig. 8, the wear rate increases with the increase of load, while it decreases with the increase of velocity despite the higher wear loss. The reasons can be deduced according to the equation, the wear rate $= \Delta v / Ld = \Delta m / \rho Ld$,

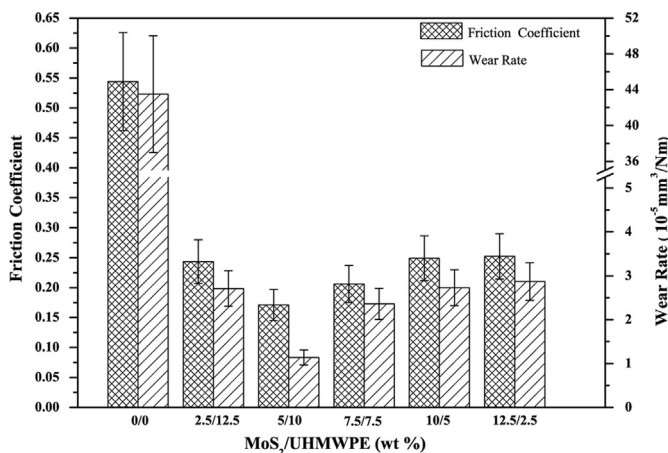


Fig. 7. Effect of mass ratio of MoS₂/UHMWPE on the friction coefficient and wear rate of TiO₂/PA 6 nanocomposite at 40 N and 200 rpm. The total solid lubricant content is constant of 15 wt%.

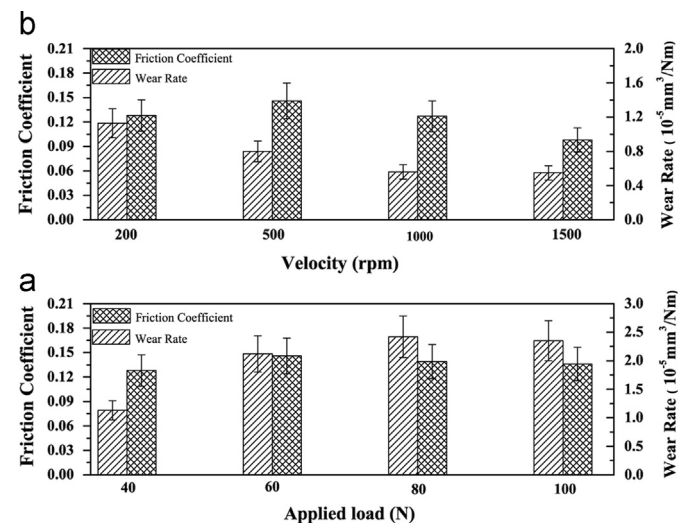


Fig. 8. Effect of load and velocity on the tribological properties of TiO₂/PA 6 nanocomposite filled with 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE, (a) effect of load at 200 rpm and (b) effect of velocity at 40 N.

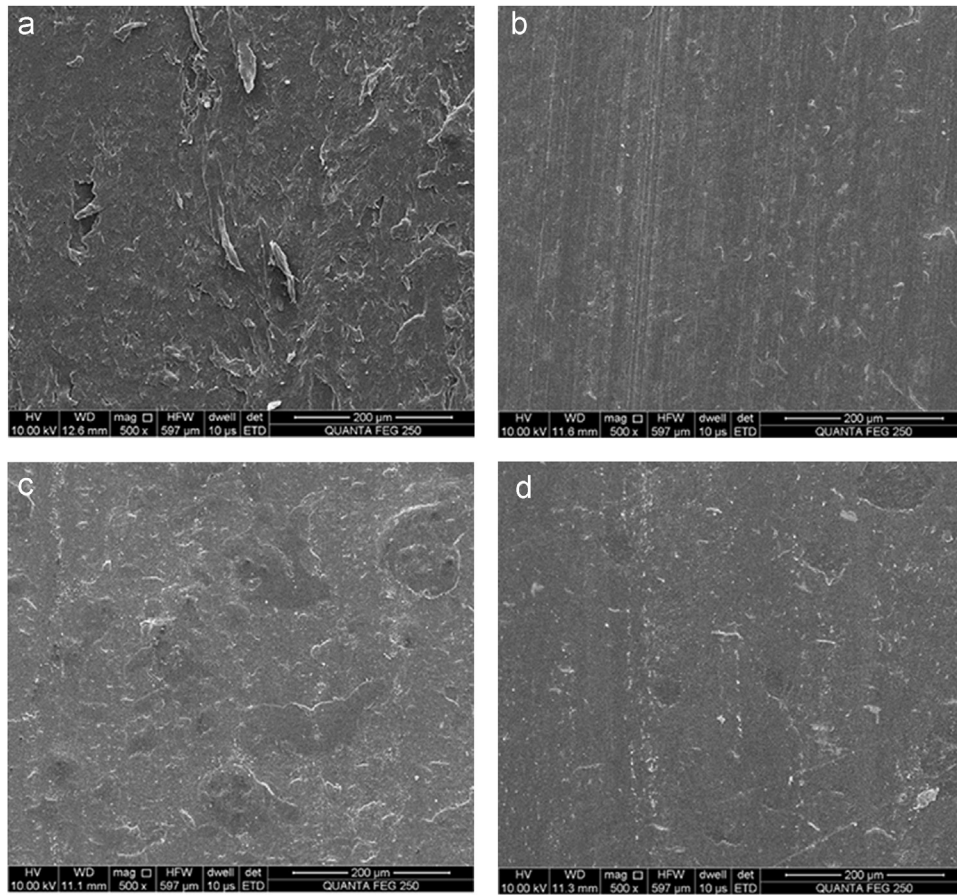


Fig. 9. SEM images of the worn surface at 40 N and 200 rpm of $\text{TiO}_2/\text{PA 6}$ nanocomposite filled with (a) 15 wt% MoS_2 , (b) 5 wt% MoS_2 + 10 wt% UHMWPE, (c) 5 wt% MoS_2 + 10 wt% PTFE and (d) 5 wt% MoS_2 + 5 wt% UHMWPE + 5 wt% PTFE.

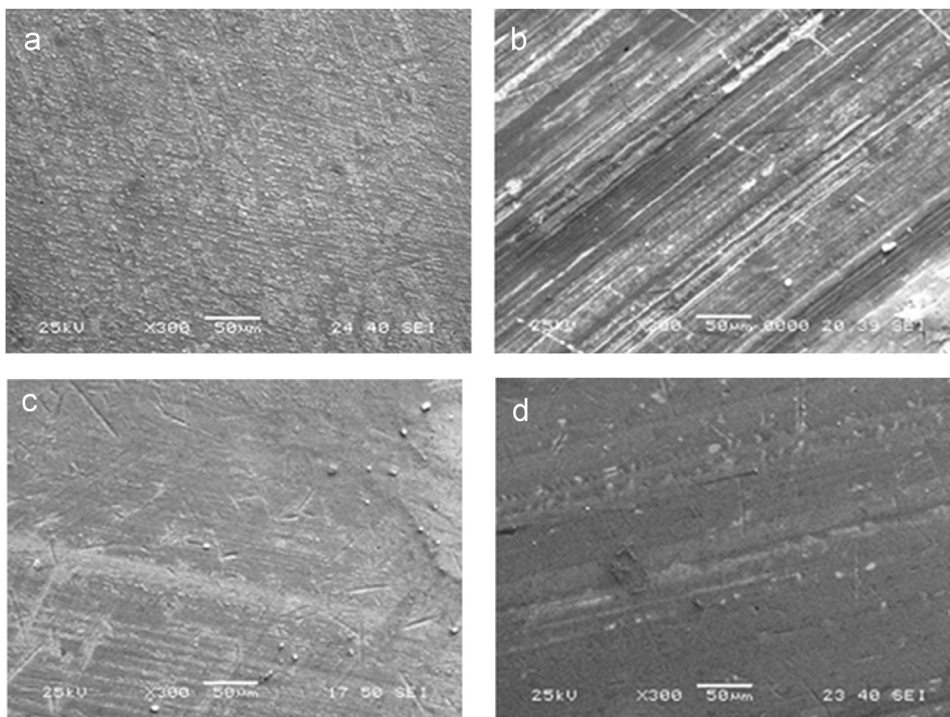


Fig. 10. SEM images of the transfer film on the counterface at 40 N and 200 rpm of $\text{TiO}_2/\text{PA 6}$ nanocomposite filled with (a) 15 wt% MoS_2 , (b) 5 wt% MoS_2 + 10 wt% UHMWPE, (c) 5 wt% MoS_2 + 10 wt% PTFE and (d) 5 wt% MoS_2 + 5 wt% UHMWPE + 5 wt% PTFE.

it can be concluded that the increased amplitude of wear loss caused by the increased load is higher than that of load; instead the increased amplitude of wear loss caused by the increased velocity is lower than that of velocity, for example, the wear rate under 500 rpm is lower than that under 200 rpm, this suggests that increasing the velocity by 2.5 times just results in an increase of wear loss by a factor lower than 2.5, consequently the wear rate decreases with the increases of velocity.

3.2. SEM and the wear mechanism studies

To have more information about the variation in wear behavior due to the addition of solid lubricants, morphologies of the worn surface and the transfer film formed on the counterface were examined by SEM. Fig. 9 shows the characteristic of the worn surfaces at 40 N and 200 rpm.

As shown in Fig. 9a, a disintegration of the top layer of the worn surface is observed in MoS₂ filled composite, the material removal takes in form of large blocks indicating the serious adhesive wear, which corresponds to the worst friction and wear behavior. In case of multiple phases of fillers filled composites, worn surfaces are much smoother, the worn surface of 5 wt% MoS₂+10 wt % UHMWPE filled TiO₂/PA 6 nanocomposite characters with mild abrasive wear in Fig. 9b, and the grinding marks are very shallow. While back transfer of the materials in patchy form on the worn surface can be seen in the micrograph of the composite filled with 5 wt% MoS₂+10 wt% PTFE as shown in Fig. 9c, which indicates adhesive mechanism; the composite filled with 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE as shown in Fig. 9d possesses the common features of both Fig. 9b and c, but the grinding grooves become more shallow and the back transfer phenomena alleviate, it can be concluded that mild adhesion and abrasive wear appear to be the main wear mechanism.

Fig. 10 shows the transfer film morphologies of MoS₂ and its combined solid lubricants filled TiO₂/PA 6 nanocomposites. As seen in Fig. 10a, a thick and discontinuous transfer film is detected for the TiO₂/PA 6 nanocomposite filled with 15 wt% MoS₂, indicating that the materials were transferred to the counterface and rubbed off easily, which confirms the highest wear rate. The morphologies of the transfer films formed by the combined solid lubricants filled composites show completely different characteristics. A fairly thick and continuous film is formed by the composite filled with 5 wt% MoS₂ + 10 wt% UHMWPE, as shown in Fig. 10b, and deep furrows which are produced on the soft and thick polymer transfer film surface due to sliding action are detected. When it comes to the transfer film formed by 5 wt% MoS₂+10 wt% PTFE filled composite, as shown in Fig. 10c, the deep grooves are no longer seen and the transfer film seems to be thinner but not so continuous; combining with the morphology of the worn surface as shown in Fig. 9c, it can be concluded that the creation and detachment of transfer film occurred within the process of sliding. The detachment speed may be greater than the creation speed, subsequently a discontinuous transfer film produced as a result. A thin, uniform and continuous transfer film is observed in Fig. 10d when 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE used as filler, it can be concluded that the FeS, FeSO₄, Fe₂(SO₄)₃ compounds increase the adhesion between the transfer film and the counterface, therefore the replenishment and the loss of the transfer film reduced which is responsible for the improved tribological properties.

4. Conclusions

The tribological properties of a series of TiO₂/PA 6 nanocomposites filled with MoS₂, UHMWPE, PTFE and their combinations

were systematically investigated. The following conclusions can be drawn:

- The tribological properties of the TiO₂/PA 6 nanocomposite were remarkably enhanced by the incorporation of UHMWPE and PTFE, whilst PTFE performs better, the best result was achieved for 5 wt% PTFE filled composite, the friction coefficient of which is 0.086 and the wear rate is $9.8 \times 10^{-6} \text{ mm}^3/\text{Nm}$. However, MoS₂ was harmful to the friction and wear behavior of TiO₂/PA 6 nanocomposite.
- MoS₂ combined with UHMWPE showed better effect than the combination of MoS₂ and PTFE; the synergistic effect between UHMWPE and MoS₂ was obtained and the optimum frictional behavior should be achieved when the mass ratio of MoS₂/UHMWPE was 5/10. Among the combinative solid lubricants filled composites, the best wear resistant was achieved by the one filled with 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE which exhibited a friction coefficient of 0.128 and a specific wear rate of $1.12 \times 10^{-5} \text{ mm}^3/\text{Nm}$.
- For 5 wt% MoS₂+5 wt% UHMWPE+5 wt% PTFE filled TiO₂/PA 6 nanocomposite, according to the scope of experimental error, the friction coefficient was not affected obviously by the load and velocity, while the wear rate increased with the increase of load but decreased with the increase of velocity.
- Tribochemical reactions between fillers and counterface played a key role on the tribological behavior. The presence of UHMWPE and PTFE facilitated MoS₂ to transform into FeS, FeSO₄, and Fe₂(SO₄)₃ which could increase the adhesion between the transfer film and the counterface. And the adhesion would retard the loss of transfer film from the counterface, as a result the wear resistance improved.

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